National Engineering College,K.R.Nagar, Kovilpatti
(An Autonomous Institution Affiliated to Anna University, Chennai)
Department of Computer Science and Engineering

23CS18E-FullStack App with Flutter and Firebase

Year/Sem: III CSE/VI 20.03.2026
Time: 9.15 am to 10.05 am

Preparatory Test-2

1. Illustrate the concept of implicit animations in Flutter by developing an interactive UI component. Create a Flutter application that displays a minimal user card with an avatar and name. When the card is tapped, it should gradually transform to reveal more information such as contact details and a short description.
   Your implementation should demonstrate:
   - Dynamic resizing of the widget
   - Smooth transition of background color
   - Gradual change in shape (border radius)
   - Appropriate use of animation timing and easing curves

   Also, explain how animation duration and curve influence the smoothness and responsiveness of UI transitions.
2. Explain how Flutter manages screen transitions using its navigation system. Compare different navigation methods such as: Adding a new screen to the stack, replacing the current screen, returning to the previous screen. Develop a multi-screen Flutter application consisting of:
   - Main Screen
   - Item Detail Screen
   - Payment Screen

   Demonstrate navigation between these screens and describe how the navigation stack evolves at each step.

3. Discuss the advantages of using named routes for organizing navigation in large Flutter applications. Create an application with:
   - A User Input Screen (collecting details like name, email, phone, etc.)
   - A Dashboard Screen

   Pass selected user information between screens using named routes and display a personalized message. Explain how named routes improve maintainability and readability.

4. Describe various scrolling widgets available in Flutter and their practical use cases. Compare Standard scrollable lists, Lazily built lists, Advanced customizable scrolling. Develop an application that displays a list of articles with:
   - Title
   - Summary
   - Date

   Use an efficient scrolling method to render at least 10 items. Enhance the UI by adding a smooth visibility effect (e.g., fade-in animation) as items appear on the screen.

5. Explain how Flutter's layout system helps in building adaptive user interfaces. Discuss the role of layout widgets such as horizontal and vertical arrangements, flexible space allocation, and container styling. Design a dashboard-style application that includes:
    - A header bar with a title
    - A row of summary cards displaying key metrics
    - A scrollable section showing recent updates

    Ensure proper alignment, spacing, and responsiveness across different screen sizes.

6. Explain how horizontal scrolling is implemented in Flutter and when to use different scrolling widgets. Develop a media browsing screen that includes:
    - A section header
    - A horizontally scrollable list of items (e.g., movies or products)
    - Each item containing an image and title

    Implement smooth scrolling behavior and enhance the user experience with a snapping or centered alignment effect.

7. Discuss how Flutter detects and responds to user input using interaction-handling widgets. Create an application that displays a grid of selectable items (e.g., categories or options). When a user selects an item:
    - It should visually indicate selection
    - Only one item should remain active at a time

    Explain how interaction widgets differ in terms of visual feedback and gesture handling.

8. Demonstrate how Flutter supports automated animations over time. Develop an application that:
    - Displays a set of profile images
    - Automatically switches between them at regular intervals
    - Uses smooth transition effects between images

    Explain how timing intervals and animation transitions affect the overall user experience.

9. Explain the concept of state management in Flutter and how Riverpod improves application scalability. Develop a simple shopping cart application that:
    - Allows adding and removing items
    - Dynamically updates total cost

    Discuss how state changes trigger UI updates and how Riverpod efficiently manages dependencies compared to other approaches.

10. Explain how Flutter applications retrieve data from external services. Create an application that:
    - Fetches data from a public API
    - Displays results in a scrollable list
    - Handles loading, success, and error states gracefully

    Describe how asynchronous state is managed and how the UI reacts to different stages of data fetching.